

NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) who builds complete AI systems solo and ships them to real users. Sole-built Trezzit (trezzit.com) — a production AI SaaS serving **600+ users** for 12+ months, iterating an LLM pipeline from **60% → 95%+** accuracy by systematically diagnosing failure modes across 4 providers. Built SiliconCrew — a 21-tool autonomous agent with self-correction, multi-provider routing, and a full MCP server — scoring **37/92 on CVDP** benchmarks with zero human intervention.

TECHNICAL SKILLS

AI & Agent Systems: LangGraph | ReAct Agents | MCP Protocol | Multi-provider LLM Routing | RAG Pipelines | Eval Pipelines | Prompt Engineering | Tool Use / Function Calling | PyTorch | CUDA

Backend & Systems: Python | C++ | FastAPI | Django REST Framework | Async Python | PostgreSQL | Redis | Celery | WebSocket | Docker | Linux

Frontend & Infra: React | Next.js | TypeScript | GitHub Actions CI/CD | DigitalOcean | AWS | Nginx | Vercel

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Arizona State University

Expected May 2026

Tempe, AZ

Coursework: Artificial Intelligence · Reinforcement Learning in Robotics · Perception in Robotics · Foundations of Algorithms · Algorithm/Hardware Co-Design

Bachelor of Engineering — Electrical and Electronics

BMS College of Engineering

August 2022

Bengaluru, India

PROJECT EXPERIENCE

SiliconCrew — Autonomous AI Agent with Eval-Driven Self-Correction

Nov 2025 – Present

- Built a LangGraph ReAct agent (21 tools) routing inference across Gemini, OpenAI, and Anthropic with provider-specific quirk handling; implemented a full MCP server (stdio/SSE/HTTP transports, auto-discovery of LangChain @tool functions) compatible with Claude Desktop, VS Code, and any MCP client; FastAPI + WebSocket backend with real-time tool-call streaming to a Next.js frontend.
- Implemented an autonomous self-correction loop — on task failure the agent reads diagnostic output, identifies root cause, patches its generation, and re-verifies without human input; achieved **37/92 first-pass** and **5/5 perfect score** on structured code generation benchmarks; async job engine with per-run isolation, exponential backoff (30s→10min), and on-disk index for process-restart recovery.
- Built full-stack observability: Next.js 3-panel client (Sidebar | Chat | Artifacts) with 6 artifact viewer tabs for inspecting agent outputs at each pipeline stage — Monaco code editor, execution trace viewer, diagram renderer, and structured report panel — WebSocket streaming of live tool-call execution cards, and per-session token and cost tracking across all LLM providers.

Trezzit — Production AI SaaS (trezzit.com)

Feb 2025 – Present

- Sole-built and shipped Trezzit (trezzit.com) — two-pass LLM receipt pipeline (pass 1: extract structured data, pass 2: semantic validation and self-correction) across a provider-abstracted registry of Gemini, OpenAI, Claude, and Grok, improving parsing accuracy from ~60% to 95%+; SHA-256 image deduplication and O(1) stored balance reads via pre-calculated pairwise debts.
- Production infrastructure: GitHub Actions CI with live PostgreSQL service container (995 tests on every push), Redis for caching and rate limiting, DigitalOcean VPS with Nginx + Gunicorn + systemd managing 4 services; PWA with VAPID push notifications, Google OAuth2, and OTP authentication — 60+ REST endpoints, 40 Django models.

HEAL.AI — Healthcare AI with Custom RAG Pipeline

Sep – Nov 2025

- Won 1st place at Devlabs Hackathon; built a custom RAG pipeline without an external vector database — PDF chunking with 2-sentence overlap, Google text-embedding-004 (768-dim) embeddings, cosine similarity retrieval with confidence scoring, and automated FDCCA-compliant dispute email generation via Gemini; FastAPI backend, React + TypeScript frontend.

CUDA MLP — GPU Architecture Coursework

Fall 2024

- Implemented 6 CUDA kernels in C++ (linear forward/backward, ReLU forward/backward) for MLP training on MNIST — compiled via nvcc to shared libraries, bound to Python through ctypes FFI, with manual cudaMalloc/cudaMemcpy buffer management.

WORK EXPERIENCE

Software Developer (Sole Engineer)

NCR GOLD

Oct 2022 – Apr 2024

Bengaluru, India

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

AWARDS & HONORS

- Won **‘Best Use of AI’** at Strategy X DevHacks’25 (ASU) — built AdaptED AI, a Gemini-powered platform generating personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Won **1st place at Devlabs Hackathon** — HEAL.AI, an AI healthcare assistant with a custom RAG pipeline for insurance Q&A and automated medical bill dispute generation.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for password generation.